INA236

User manual

DC current and voltage measurement module

ESP32 · Arduino IDE · I2C

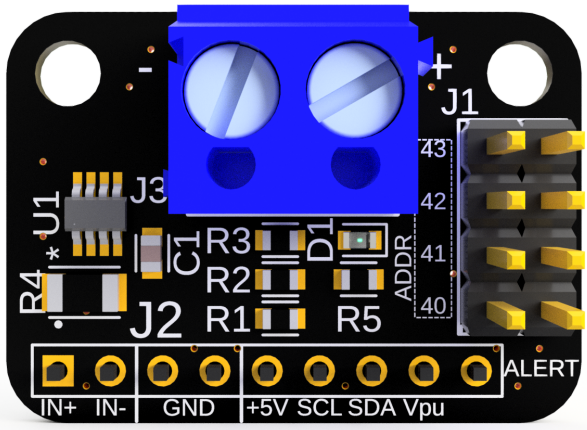


Figure 1. Front view of the module. Image supplied by the designer.

Two connection paths, two calibration values.
Input A: lower header, 25 mΩ, 0 to 3.2 A.
Input B: blue terminal block, 31 mΩ equivalent, calculated 0 to 2.58 A.
Measurement bus voltage: 0 to 48 V DC.

Before connecting

The pin marked +5V accepts a 3.3 V or 5 V supply. Vpu sets the I2C pull-up voltage: use 3.3 V with ESP32. The 48 V rating applies to the measured circuit, never to the supply or digital pins.

Version 1.0 · October 16, 2026 · English

IMPORTANT BEFORE CONNECTING
Blue terminal block: the physical board has no + or - symbols on its silkscreen. In the orientation of Figure 1: **right = IN+; left = IN-**. Symbols in the image are visual references only.
Common ground is required: connect measured-circuit GND, INA236 GND and ESP32 GND together. Without this shared reference, voltage will not be measured correctly. See page 2 for details.

01 / Connections and wiring

9-pin J2 header: manual reference numbering, left to right when viewing the module as shown in Figure 1.

Pin	Terminal	Function / connection
1	IN+	Positive measurement input A; toward positive source.
2	IN-	Negative measurement input A; toward load positive.
3	GND	Common ground for supply, microcontroller and measured circuit.
4	GND	Second common ground connection.
5	+5V	Module supply: 3.3 V or 5 V.
6	SCL	I2C clock. ESP32: GPIO22.
7	SDA	I2C data. ESP32: GPIO21.
8	Vpu	I2C pull-up supply: 3.3 V for ESP32; 5 V only for compatible logic.
9	ALERT	Alert output. ESP32: GPIO27; external 4.7 kΩ pull-up to 3.3 V.

Current path: select A or B

Source (+) → IN+ → module → IN- → load (+).
Source (-) → load (-) and module/ESP32 GND.

For A, use pins 1 and 2. For B, use the blue terminal block: in Figure 1 the right terminal is (+)/IN+ and the left terminal is (-)/IN-. These are alternative paths through the same sensor, not two independent channels. Use one path at a time.

Wire with all supplies off. IN- goes toward the load and is not the GND return. Do not connect IN+ and IN- directly across the source positive and negative terminals: the shunt has very low resistance. The module provides no galvanic isolation.

NOTICE: THE TERMINAL BLOCK HAS NO + / - MARKINGS
The + and - symbols shown in Figure 1 **are not printed on the physical board silkscreen**. The orientation stated in this manual is correct: view the module from the front as in Figure 1, with the blue terminal block at the top and the 9-pin header at the bottom. The **right terminal is IN+ (+)** and the **left terminal is IN- (-)**. Identify the terminals using this orientation before wiring.

REQUIRED: COMMON GROUND FOR VOLTAGE MEASUREMENT
Connect **measured-circuit GND, INA236 GND and ESP32 GND together**, even when powered from separate supplies. In the high-side connection described here, source/load negative is the measured-circuit GND.
The INA236 measures bus voltage between **IN- and GND**. Without a shared ground, the voltage reading will be incorrect or undefined. **IN- does not replace the GND connection**: in this setup, IN- connects to load positive.

02 / Ranges and calibration

The sketch sets ADCRANGE = 0 (±81.92 mV). Current is derived from I = Vshunt / R. The bus range is 0 to 48 V DC; bus voltage is measured from IN- to GND [1].

Path	Equivalent R	Use limit	Full scale[1]
A / Header	25 mΩ	3.20 A	≈ 3.277 A
B / Terminal block	31 mΩ	2.58 A[2]	≈ 2.643 A

Calculation for input B

At 3.2 A, input A produces 3.2 × 0.025 = 0.080 V (80 mV). Keeping the same drop for B:
I_B = 0.080 / 0.031 = 2.58065 A → use up to 2.58 A.
The theoretical ADC limit is 0.08192 / 0.031 = 2.64258 A.

[1] Approximate values at nominal ADC range end; the largest positive code is 81.9175 mV. Do not use full scale as an operating margin.
[2] Calculated using the same 80 mV drop, not a thermal qualification of the board. Continuous capability also depends on the shunt, traces, connectors and temperature; module thermal characterization is not available.

Select resistance in the program

Edit **SHUNT_RESISTANCE_OHM** before compiling [2]:

Input	Sketch value
A / 25 mΩ	0.025f
B / 31 mΩ nominal	0.031f
B / current bench adjustment	0.031107383f

The reviewed local copy uses B = 31.107383 mΩ, a one-point adjustment against a 298 mA reference. This is not an independent resistance measurement or a guarantee of accuracy across the range. At this value, 80 mV corresponds to 2.57173 A: use up to **2.57 A**; full scale is ≈ 2.63346 A.

Using 25 mΩ for an actual 31 mΩ path overstates current by about 24%; using 31 mΩ for the 25 mΩ path understates it by about 19.35%. The program recalculates calibration and alert threshold; the constant applies to all detected sensors.

03 / Startup and support

1. Download the sketch from repository [2]. In Arduino IDE, install ESP32 board support and select the appropriate ESP32 board and port. The sketch uses Arduino.h and Wire.h provided by that environment.

2. Open sketch_INA236.ino and select the A or B resistance from the previous page. Check the wiring on page 2. For ESP32, connect Vpu and the ALERT pull-up to 3.3 V, even if the module is powered at 5 V.

3. With supplies off, select one address using ADDR/J1 (0x40, 0x41, 0x42 or 0x43). Use 0x40 to test the default alert. Do not bridge several options at once; verify the detected address at startup.

4. Upload the program, open Serial Monitor at 115200 baud and reset. The sketch checks the four addresses at 100 kHz; confirm OK and System initialized.

5. Apply a small, steady DC load. Check positive current, bus voltage and agreement with a reference instrument. Increase gradually without exceeding the selected input limit.

Alert and readings

The default alert is for positive current above 0.500 A, on 0x40 only. ALERT is open-drain and active LOW; GPIO27 needs the specified pull-up. It is an indication signal and does not disconnect the load. Readings print approximately every 200 ms. Current and Calculated current use the same ADC; agreement does not independently establish accuracy.

Troubleshooting

NOT FOUND / I2C ERROR: check power, common ground, SDA/SCL, Vpu and address selector; reset after correcting.
Current error: check A/B path and configured resistance; verify zero and several steady currents.
Negative current: check IN+ → IN- direction.
Math overflow = YES or clipped reading: reduce current and check configuration. NO does not rule out ADC saturation.

References and scope

[1] Texas Instruments. INA236, SBOSA81D, Rev. D; ranges, registers and interface.
https://www.ti.com/lit/ds/symlink/ina236.pdf
[2] ErickParraFlores. INA236_Sketch; code for ESP32 / Arduino IDE.
https://github.com/ErickParraFlores/INA236_Sketch

Software basis: local sketch_INA236.ino and calibration notes; revision 81f93aa99316, reviewed October 16, 2026. The repository may change. Module pinout and limits were supplied by the designer; B calculations are documented here. The image is a visual reference, not a mechanical drawing.